

AYE
TECH HUB

• COOHOM INTERIOR DESIGN • LESSON 03

Space Planning Basics

Plan for Function and Flow Before Placing Anything

Lesson 03 of 30 | Coohom Interior Design Program

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- The 7 principles of good space planning every designer must know
- Minimum clearance standards for every room type (with measurements)
- Traffic flow: primary, secondary and tertiary circulation paths
- Zoning: how to divide a space into activity areas that work together
- Practical space planning exercises for living room, bedroom, kitchen

Space Planning — The Foundation of Great Design

Space planning is the process of arranging a room's elements — furniture, circulation paths, activity zones — to create a space that **works efficiently for the people who live or work in it**. The biggest mistake most beginners make is jumping straight into placing furniture and choosing colours before thinking about how the space will actually be used day to day.

Professional interior designers always start with a space plan. In Coohom, you work in 2D Floor Plan view (F1) first — arranging everything logically — then switch to 3D (F2) to see how it feels. Getting the space plan right saves hours of redesign later.

The 7 Principles of Good Space Planning



Figure 1 — The 7 Principles of Good Space Planning

Principle	The Question to Ask	Common Mistake Without It
Function First	What activities happen here daily? At what times? By how many people?	Beautiful room that does not work — no good reading light, nowhere to put things
Traffic Flow	Can people move naturally from door to door without squeezing past furniture?	Furniture blocks natural paths — people feel cramped immediately on entering
Focal Point	What is the first thing someone's eye goes to when entering the room?	Visual chaos — eye has no place to rest, room feels restless and cluttered
Balance	Does the visual weight feel even? Or is one side heavy and the other empty?	Room feels lopsided — all large pieces on one wall, empty void on the other
Scale & Proportion	Is the furniture the right size for the room and for each other?	Oversized sofa dominates small room; tiny rug "floats" in large space
Natural Light	Where does light enter? How does it change through the day?	Seating placed with sun in occupant's eyes; dark work area in shadow
Flexibility	Can the space serve more than one purpose when needed?	Single-use room is useless when that use changes — dead space investment

Clearance Standards and Traffic Flow

Clearance standards are the minimum distances between furniture pieces, walls, and circulation paths that make a room comfortable and safe to use. These are not rules of style — they are ergonomic requirements based on human body dimensions. Violating them makes a space feel cramped and frustrating even when it looks good in a render.

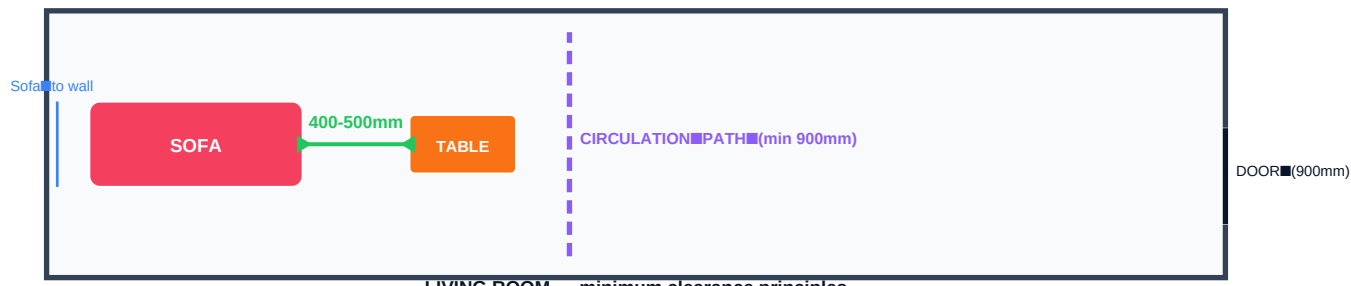


Figure 2 — Minimum Clearance Standards Applied to a Living Room

Essential Clearance Standards by Room Type

Space / Clearance	Minimum	Comfortable	Notes
Primary circulation (main walkway)	900 mm	1 200 mm	Path between doors; wheelchair users need 900mm minimum
Secondary circulation (around furniture)	600 mm	750 mm	Space to walk past seating, alongside beds
Sofa to coffee table	350 mm	450 mm	Close enough to reach items, far enough to walk past
Dining chair pull-out space	750 mm	900 mm	From table edge to wall — allows standing from chair comfortably
Bed to wall / other furniture	500 mm	700 mm	One side can be 300mm if against a wall, but primary side needs 500mm+
Kitchen work triangle (total perimeter)	3 600 mm	6 000 mm	Fridge-Sink-Hob. Too large is as inefficient as too small.
Bathroom vanity clearance	700 mm	900 mm	From vanity front to wall/toilet — allows standing and bending
Door swing clearance	Full swing + 300mm	Full swing + 600mm	Door must open fully without hitting furniture

Traffic Flow — Primary, Secondary and Tertiary Paths

Path Type	What It Is	Clearance	Example
Primary circulation	Main entry and exit routes through the space — used by everyone always	Min 900 mm	Front door → kitchen; living room entrance → all seating areas
Secondary circulation	Routes used to access furniture or areas — used frequently but not always	Min 600 mm	Around dining chairs; beside beds; into wardrobe area
Tertiary circulation	Occasional access routes — used rarely, acceptable to squeeze slightly	Min 450 mm	Behind a chair that is rarely moved; corner access for cleaning

The "float test": In Coohom's 2D floor plan view, trace every path a person would take in a normal day — morning routine, cooking, watching TV, having guests. If any path is blocked or requires squeezing between

objects, you have a circulation problem to solve before adding a single cushion or colour.

Zoning, Scale and Practical Application

Zoning is the process of dividing a space into defined areas based on activity. In open-plan layouts — the most common in modern design — zoning is what prevents the space from feeling like one undifferentiated room. Zones can be created without walls using furniture arrangement, rugs, lighting, and ceiling treatments.

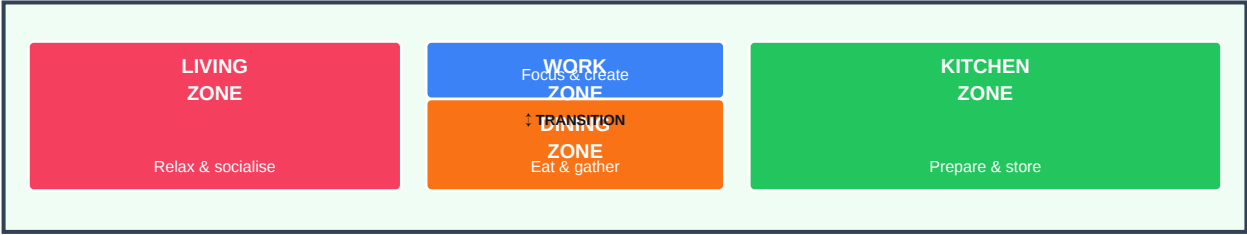


Figure 3 — Activity Zoning: Dividing a Space by Function

Zoning Tool	How It Defines a Zone	Coohom Feature Used
Furniture arrangement	Sofas facing inward define a conversation zone; back to back defines separation	Furniture placement in 2D view; use sofa/chair groupings
Area rugs	Rug anchors a seating or dining group — everything on the rug = one zone	Flooring material per zone (apply different floor finish to each area)
Lighting	Pendant over dining table, floor lamp in reading corner define distinct zones	Place lighting fixtures per zone; use different colour temperature settings
Level change	A step up or down, or a raised platform, creates a clear zone boundary	Use slab or platform element at different Z-height for raised areas
Ceiling treatment	Coffered ceiling, dropped ceiling, or change of material above each zone	Add feature ceiling above each zone using ceiling tool
Colour / material	Different wall colour or accent material behind zone groups elements visually	Apply different wall finish/colour to walls of each zone

Room-by-Room Space Planning Checklist

Room	Start With	Key Clearance to Check	Coohom Tip
Living Room	Sofa position relative to focal point (TV/fireplace)	900mm circulation; 400mm sofa-to-table; 600mm TV-to-sofa minimum	Draw room in 2D first; place sofa, then arrange other seating around it
Master Bedroom	Bed position (usually centred on main wall, not corner)	500mm each bed side; 700mm wardrobe access; clear door swings	Place bed first — everything else follows. Use king/queen bed template.
Kitchen	Work triangle: Fridge-Sink-Hob within 3.6-6m total	1200mm between opposing counters (1000mm absolute min)	Use kitchen cabinet modules; verify work triangle in 2D before 3D
Dining Room	Table size and shape relative to room and number of seats	900mm clearance all around table; 750mm chair pull-out	Place table with correct chair count; verify pull-out zone is clear
Home Office	Desk facing wall or window — not facing a distraction	900mm behind chair; clear path to door; cable management space	Position desk for natural light from side (not behind screen)
Bathroom	Vanity and toilet positions first (fixed plumbing points)	700mm vanity clearance; 600mm toilet side clearance; shower access	Sanitary items are fixed — design around them, not the other way

What comes next — COHOOM-004: Floor Plan Creation — drawing your first complete room from scratch in Coohom: setting room dimensions, drawing walls accurately, adding doors and windows, and setting up rooms ready for furniture placement.